

Appendix C: replay transcripts and expected markers

Repository source: proof/APPENDIX_C_REPLAY_TRANSCRIPTS.md

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Status: summary of archived and transferred evidence **Construction-transcript authority:** checksummed files under `certificate/transcripts/` **Remote-evidence index:** `certificate/evidence/rc1_remote_audit.json`

This appendix does not replace the transcript or log files. It identifies what each record contains, what a successful run must show, and how the evidence should be interpreted.

C.1 Construction transcript inventory

File	Path exercised	Required terminal marker
<code>exact_certificate.txt</code>	Production exact-rational verifier	MINVOL UNIVERSAL CONTACT-BOUND CERTIFICATE: EXACT
<code>binary64_audit.txt</code>	Independent NumPy binary64 audit	MINVOL UNIVERSAL CONTACT-BOUND BINARY64 AUDIT: PASS
<code>python_unittest.txt</code> <code>adversarial_mutations.txt</code>	Ten reconstruction/regression tests Seven deliberate corruptions	Ran 10 tests followed by OK MINVOL ADVERSARIAL MUTATION SUITE : PASS
<code>base_r_audit.txt</code>	Independent base-R audit	MINVOL UNIVERSAL CONTACT-BOUND BASE-R AUDIT: PASS
<code>export_equivalence.txt</code>	Private-source/public-export theorem-field comparison performed during construction	PUBLIC EXPORT EQUIVALENCE: PASS
<code>environment.txt</code>	Construction-environment versions and platform	no theorem marker; provenance only

Exact SHA-256 values for these files are recorded in `certificate/SHA256SUMS.txt` and the root `SHA256SUMS.txt`.

C.2 Exact transcript

The exact transcript should report, among other values:

<code>split radius:</code>	0.61139120200000000000
<code>weakest lower row:</code>	03B02B
<code>changed fan vertices:</code>	18229
<code>minimum exact fan margin:</code>	0.0000000002888904207092857876733697822441644655792123744
<code>near-Jung handoff margin:</code>	0.0000000000025460541895664424549959552655663257281256941

```
universal coefficient lower: 0.4118796712122863775432315385981549188973081955671703077
```

The exact output is a human-readable projection of integer and rational computations. The production program also requires byte-identical reconstruction of the authoritative certificate and exact audit artifacts.

C.3 Binary64 transcript

The numerical audit should reconstruct the same geometry with ordinary floating-point operations and report values agreeing at the documented scale. It is expected to differ in the last displayed digits from exact decimal renderings.

A pass means the independent numerical implementation agrees within its stated tolerances. It is not a substitute for the exact finite proof.

C.4 Base-R records

The archived construction transcript records a successful base-R audit. The later remote fresh-clone transcript records the exact named-author environment:

```
Rscript (R) version 4.6.1 (2026-06-24)
```

The hosted R job also installed R 4.6.1 and ended with:

```
MINVOL UNIVERSAL CONTACT-BOUND BASE-R AUDIT: PASS
```

The program reports the four refined rows, fan census and minimum margin, determinant lower bound, coordinate widths, cap totals, disjointness gap, near-Jung value, theorem coefficient, and predecessor gain.

C.5 Mutation transcript

A successful mutation transcript shows seven tests and ends with:

```
Ran 7 tests
OK
MINVOL ADVERSARIAL MUTATION SUITE: PASS
```

The test succeeds only when every corrupted object is rejected. A mutation that passes through the verifier is a test failure, even if the final coefficient is numerically unchanged.

C.6 Export-equivalence transcript

The standalone public package was compared semantically with the authoritative private theorem package during construction. Ten theorem-critical fields were checked, including the coefficient, split, lower rows, bottleneck, fan minimum, core, cap total, near-Jung value, and handoff margin.

This transcript records construction provenance. The public release does not require access to the private repository to replay its own proof.

C.7 Hosted and fresh-clone evidence

The audited release-candidate commit is

```
892d4bc6fad07950e78da66e45b95063a6415af6
Add release-candidate citation and licensing metadata
```

The successful hosted workflow is <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>. It contains three jobs: exact/Python verification, independent base R, and paper/document build plus Poppler preflight.

The named-author fresh-clone transcript records Python 3.14.7, NumPy 2.3.5, R 4.6.1, the complete replay pass marker, a ten-page paper build, 22 committed and 22 rebuilt Markdown/PDF pairs, an empty final `git status --short`, and a complete five-commit bundle.

The transferred evidence artifacts have these SHA-256 values:

2c324230f02ba4cb2541e0d353cc7906ae460cac0a768267144f650a0278a1a3	minvol-degree3-contact-bound-v1.0.0-rc1
-main.bundle	
ce2abd06646fdf89d6e60b9069ee55451715a35970afedd24e4acf0d1d64de7f	minvol-degree3-contact-bound-v1.0.0-rc1
-fresh-replay.txt	
44fc68b36e05153fabd2a93aaa4b092b946016b9eb10bcecf37d7bf5c6b4ad3a	minvol-degree3-contact-bound-v1.0.0-rc1
-github-actions-logs.zip	

The transfer manifest itself has SHA-256 3f958520e0e119cad4db97d60a2ac3702b8144a8f6228068bdf6d43124f57d. The action-log ZIP has 36 entries and passes ZIP integrity testing.

These raw artifacts are intended for the final replay-transcript release asset; they are not theorem proof objects. Their machine-readable evidence summary is `certificate/evidence/rc1_remote_audit.json`.

C.8 Reproducing the records

Run the Python paths:

```
./run_python_checks.sh
```

A successful Python run now also includes:

```
MINVOL REMOTE-EVIDENCE CHECK: PASS
```

Run the independent R path:

```
./run_r_checks.sh
```

Run both language paths:

```
./run_all.sh
```

To refresh a transcript deliberately, capture the program output, review it, refresh checksum manifests, and commit the source and new record together. Never edit a transcript to manufacture a pass marker.

C.9 Interpretation

The transcript set demonstrates repeatability of the finite programs on the recorded environments. Hosted CI and the named-author fresh clone broaden the platform evidence but do not demonstrate:

- external subject-matter review;
- novelty;
- correctness of every mathematical reduction outside the finite checker;
- immunity to a common bug shared by multiple implementations;
- sharpness or a minimizing body; or
- peer-reviewed publication.

Those distinctions must remain visible in any release announcement.